

PORTFOLIO CASE STUDY

Regression modeling • Efficiency residuals • Cap/floor simulations

Built as a reproducible R project for baseball analytics review

MLB PAYROLL & COMPETITIVE BALANCE

How much does money buy wins?

Advanced R analysis of team payroll, wins, diminishing returns, and salary-cap / salary-floor simulations using 2011–2024 MLB team-season data.

420

team-seasons

2011–2024

analysis window

Executive answer

Payroll helps, but it does not fully determine winning.

The advanced version separates three ideas that often get blurred together: spending level, roster structure, and policy compression.

14.0%

variation explained by relative payroll alone

43.8%

variation explained by the full roster/year model

46.1%

predicted SD reduction under a 0.75x–1.25x band

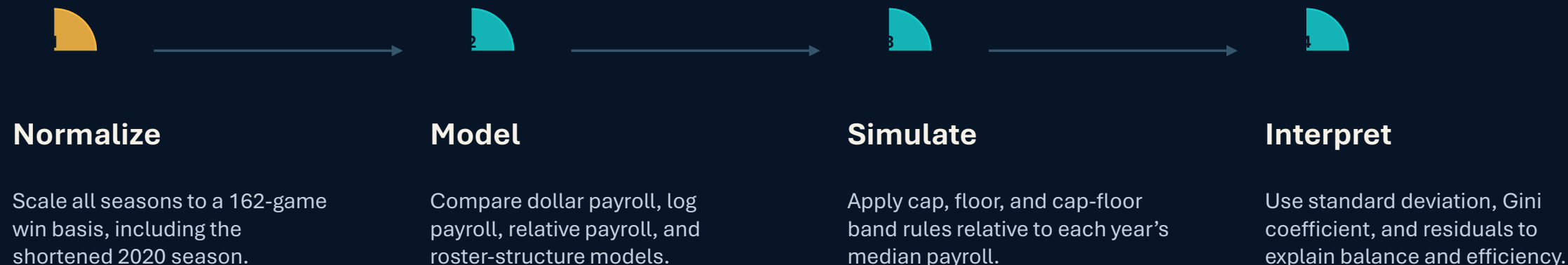
Higher payroll is associated with more wins, but the relationship is moderate.

A 10% increase in relative payroll is associated with about 1.1 additional 162-game wins in the payroll-only relative model.

Compressing only the top of payrolls helps; combining a cap with a floor compresses predicted outcomes much more.

Research design: from baseball question to model

The project is structured to answer the question rather than just produce a chart.



Key idea: model payroll as a continuous driver, then ask how the predicted win distribution changes when the spending distribution is compressed.

Data and feature engineering

Item	Design choice
Observations	420 team-seasons: 30 MLB teams × 14 seasons
Outcome	Wins scaled to a 162-game season using win percentage
Payroll scale	Total payroll allocations converted from dollars to numeric values
Inflation control	Relative payroll = team payroll ÷ season median payroll
Roster structure	Active, injured, retained, and buried payroll shares
Policy unit	Caps and floors defined relative to each season's median payroll

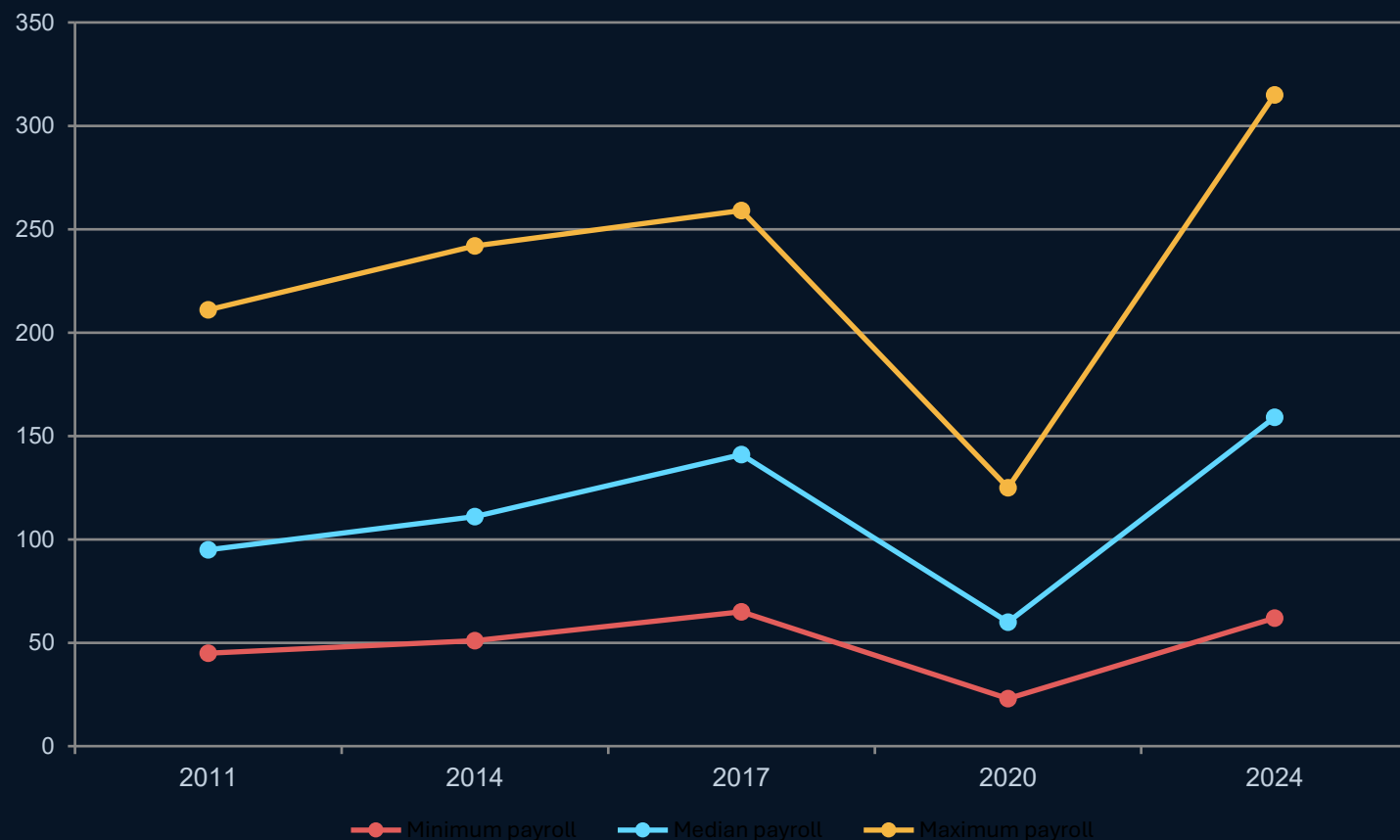
Why adjust 2020?

Raw wins are not comparable across a shortened 60-game season and normal 162-game seasons. Scaling by win percentage keeps the outcome on the same baseball scale across all years.

Data source note

The analysis uses a Kaggle dataset of MLB team payrolls from 2011–2024; payroll information in that dataset is sourced from Spotrac. The raw CSV should be downloaded from the original source when reproducing.

Payroll gaps are large and persistent



2024 snapshot

\$315M

highest payroll

\$159M

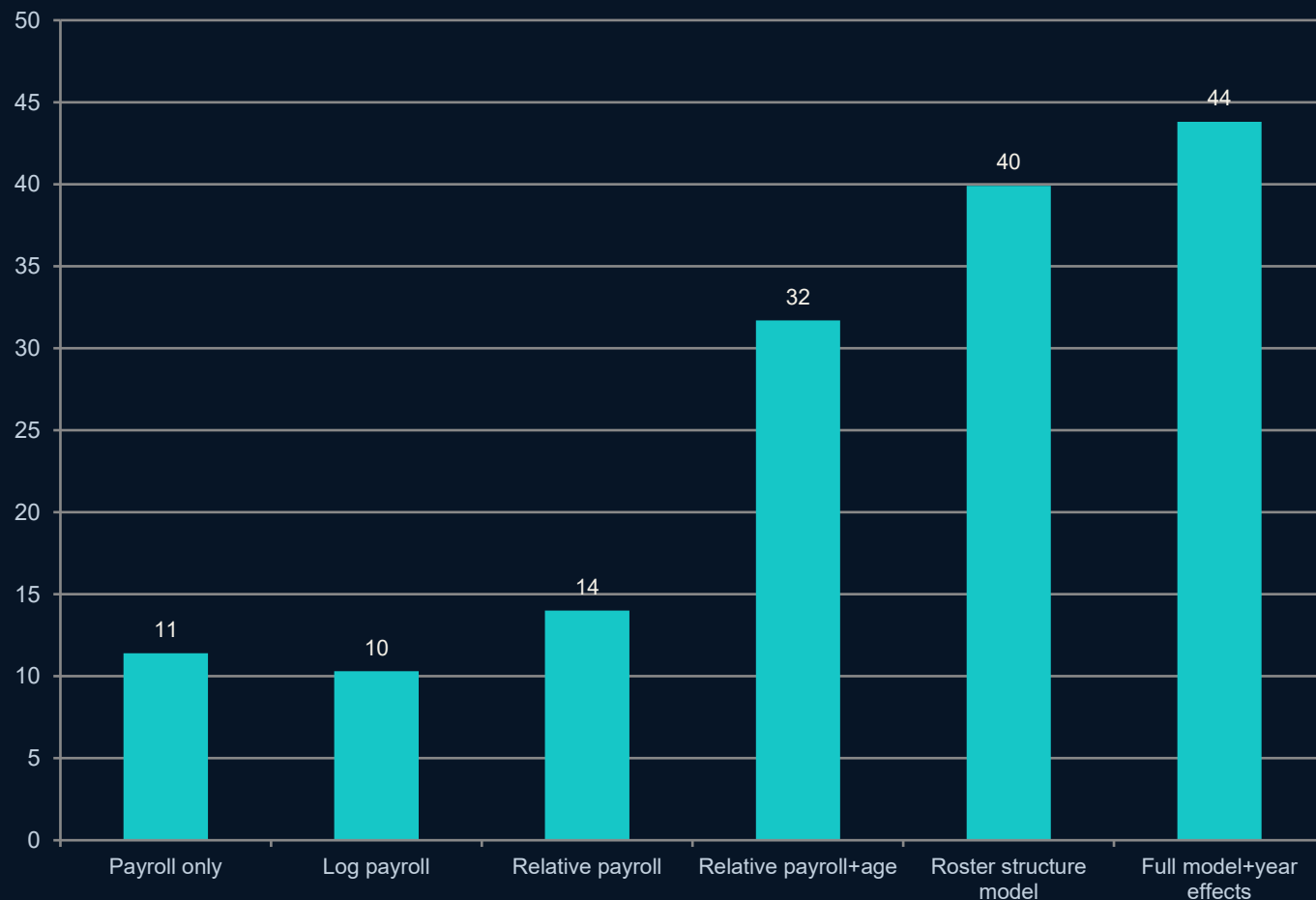
median payroll

\$62M

lowest payroll

The simulation rules are relative to each season's median, so a 2011 cap and a 2024 cap adjust naturally to league payroll growth.

Payroll matters — but it is not a complete model



What changes in the advanced version?

Uses relative payroll to compare teams within each season's financial environment. Adds roster structure features rather than treating every dollar as identical. Adds season controls in the full model to avoid confusing time effects with team effects.

Important interpretation

Even the fuller model leaves substantial unexplained variation. That is the baseball part of the result: money creates advantages, but execution, injuries, player development, and randomness still matter.

Diminishing returns: more money helps, but not evenly

Quartile	Avg. relative payroll	Avg. wins	Playoff rate
Q1	0.58×	76.4	24.8%
Q2	0.87×	77.8	24.8%
Q3	1.12×	82.3	33.3%
Q4	1.58×	87.6	54.3%

Quartile interpretation

Lowest payroll quartile

76.4

avg. 162-game wins

Highest payroll quartile

87.6

avg. 162-game wins

The gap is meaningful, but the top quartile is not automatic dominance — the spread within each group is why modeling residuals matters.

Reading the table

Higher-spending teams win more on average, but the increase is gradual rather than automatic. That is the practical meaning of diminishing returns in this project.

The advanced model separates spending from roster structure

Instead of asking only “how much did a team spend?”, the upgraded model asks where payroll was allocated and what roster profile came with it.

Feature family	Why it matters
Relative payroll	Controls for the financial environment within each season
Average age	Proxy for veteran/young roster composition
Active share	Payroll actually on the active roster
Injured share	Payroll tied up in injured players
Retained share	Payroll paid to players no longer producing for the club
Year effects	Controls for league-wide season differences and payroll inflation

43.8%

R^2 in full model

9.5

RMSE in wins

This is not a causal claim. It is a stronger descriptive model that better explains the historical data.

Efficiency lens: who beat the payroll model?

Residuals show the difference between actual 162-game wins and wins predicted from payroll with season context.

Top overperformers	Residual	Postseason
2020 TB	+35.4	Div. Winner
2023 BAL	+29.3	Div. Winner
2023 TB	+26.2	Wildcard
2020 LAD	+25.6	Div. Winner
2021 TB	+25.4	Div. Winner

Top underperformers	Residual	Postseason
2024 CHW	-38.3	No
2018 BAL	-33.8	No
2019 DET	-32.2	No
2021 ARI	-25.9	No
2022 WSH	-25.2	No

Why this matters for baseball analytics

The most interesting teams are often not the richest teams — they are the teams that turn limited payroll into more wins than the model expects.

Counterfactual policy simulation

The simulation does not claim to predict real roster moves. It asks a narrower modeling question: if payrolls were mechanically compressed, how would predicted win dispersion change?

Cap rule

$$\text{Payroll}^{\text{cap}} = \min(\text{Payroll}, \text{cap} \times \text{median}_Y)$$

Floor rule

$$\text{Payroll}^{\text{floor}} = \max(\text{Payroll}, \text{floor} \times \text{median}_Y)$$

Cap-floor band

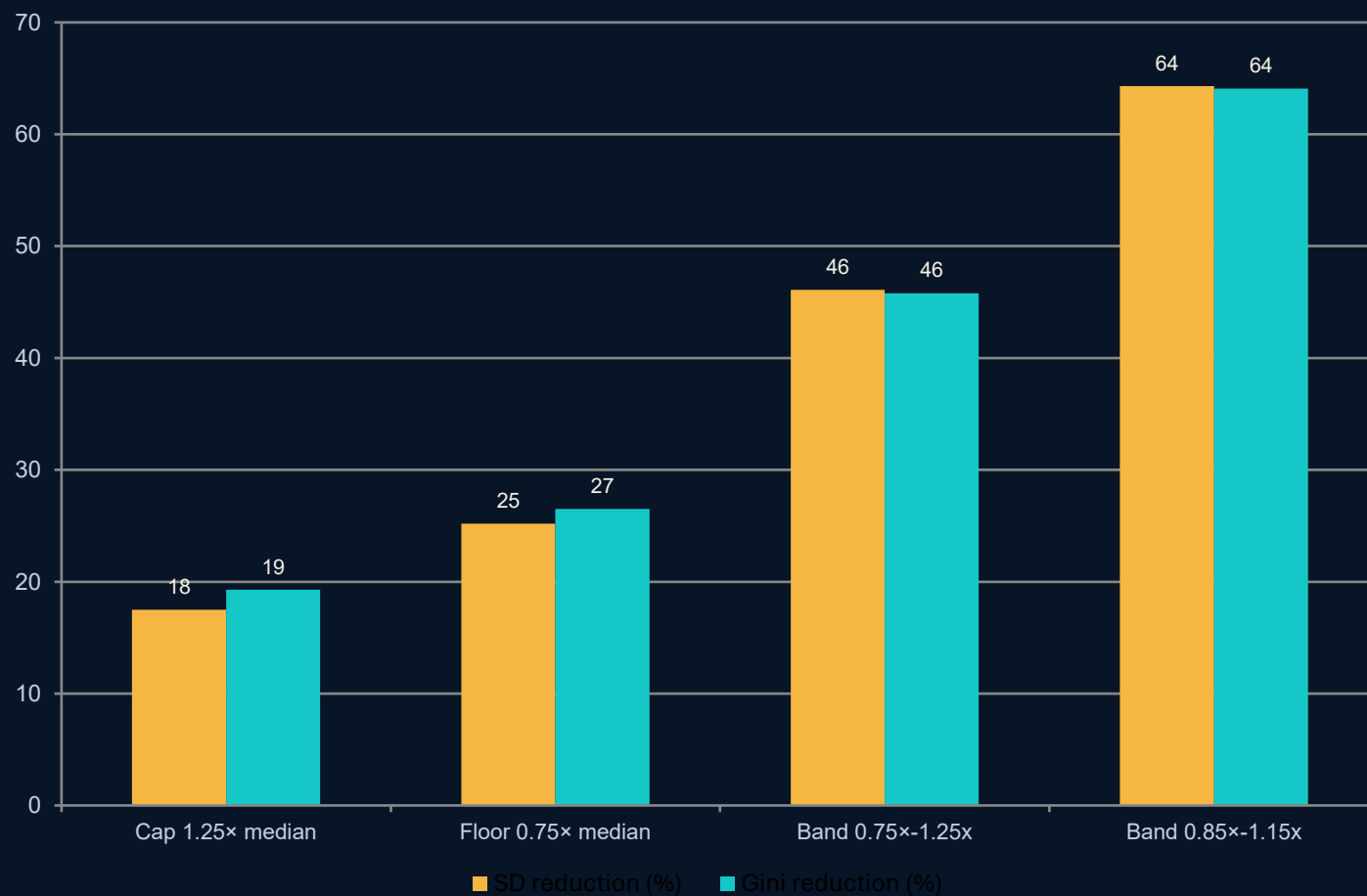
$$\text{Payroll}^{\text{band}} = \min(\max(\text{Payroll}, \text{floor}), \text{cap})$$

Balance metrics

Standard deviation of
predicted wins

Gini coefficient of
predicted wins

A cap-floor band changes the model more than a cap alone



Best practical comparison

46.1%

SD reduction for 0.75x-1.25x band

224

affected team-seasons

Interpretation

A cap can limit the top, but a floor also raises the bottom. That is why the band compresses predicted wins more than the cap alone.

Pirates context: smaller-payroll application

This slide is not a team evaluation. It shows how the model can frame a practical front-office question: when payroll rank is near the bottom, where can wins above expectation come from?

Year	Payroll rank	162-game wins	Win rank	Residual
2021	28/30	61.0	27/30	-10.6
2022	28/30	62.0	27/30	-8.4
2023	28/30	76.0	21/30	+3.3
2024	29/30	76.0	23/30	+2.2

Portfolio value

The project moves beyond “big payrolls win” and creates a reusable workflow for identifying overperformance, underperformance, and policy sensitivity.

That is the part most relevant to baseball analytics work.

Recent Pirates residuals moved from negative to modestly positive in 2023 and 2024, meaning actual wins were above the payroll-model baseline in those seasons.

What the model does not prove

The project is meant to be a quantitative framework, not a complete explanation of baseball performance.

Correlation $\neq$ causation

The model estimates historical associations, not causal effects of adding payroll.

Payroll is not talent

A better model would include WAR, run differential, player projections, contracts, and injury history.

Policy response is simplified

Real teams, agents, and players would change behavior under a new cap or floor.

Data licensing

The raw CSV should be referenced back to its source instead of redistributed without checking the license.

Best next version

Merge payroll with player WAR, injury days, roster age curves, and team development metrics to separate “spending more” from “spending efficiently.”

Reproducible portfolio package

What the reviewer should see

A clear baseball question that leads to measurable outputs.
A reproducible R script, not just a presentation.
Multiple models with honest interpretation of model limits.
Policy simulations that answer “what changes if the payroll distribution changes?”
A path to a more advanced baseball operations model.

Final deliverables

mlb_payroll_competitive_balance.R

Full analysis script: cleaning, feature engineering, regression, residuals, policy simulations, output tables, and charts.

MLB_Payroll_Competitive_Balance_Advanced_Portfolio.pptx

Professional slide deck designed to explain the project to a technical or baseball audience.

Bottom line: money matters, efficiency matters more than the simple payroll story, and payroll compression changes predicted balance most when both the top and bottom are addressed.